1. Dataset Selection & Description

This report documents a comprehensive text classification project utilizing machine learning techniques to classify documents into six distinct categories. The dataset used in this problem is a dataset from kaggle, Available at:https://www.kaggle.com/datasets/parulpandey/emotion-dataset. Three classification algorithms (MultinomialNB, LogisticRegression, and LinearSVC) were trained and evaluated on a dataset of 2,000 test samples. **Linear Support Vector Machine emerged as the superior model with 90.40% accuracy**, significantly outperforming the baseline Naive Bayes classifier (79.75%).

2. DATASET & METHODOLOGY

2.1 Data Overview

- Training Set: 16,000 samples (combined training + validation)
- Test Set: 2,000 samples
- Features: Text documents with multi-class labels
- Classes: 6 categories with imbalanced distribution

Class Distribution (Training Data):

Class	Count	Percentage
0	4,666	23.3%
1	5,362	26.8%
2	1,304	6.5%
3	2,159	10.8%
4	1,937	9.7%
5	572	2.9%

2.2 Preprocessing Pipeline

1. Text Normalization: Conversion to lowercase
2. URL Removal: Elimination of hyperlinks using regex
3. Special Character Removal: Retention of alphabetic characters and whitespace only
4. Whitespace Normalization: Removal of extra spaces and trim operations
5. Stop Word Removal: Exclusion of 179 common English words
6. Tokenization: Break text into individual word tokens

Output: **20,000 cleaned documents with vocabulary size of 4,891 features**

2.3 Vectorization: TF-IDF

Term Frequency-Inverse Document Frequency (TF-IDF) was employed to convert text into numerical features:

- min_df: 3 (minimum document frequency)
- max_df: 0.9 (maximum document frequency)
- ngram_range: (1,2) (unigrams and bigrams)

Vocabulary Size: **4,891 unique terms**

TF-IDF emphasizes domain-specific vocabulary while downweighting common terms, making it ideal for text classification tasks.

3. MODEL TRAINING & EVALUATION

3.1 Classification Algorithms

Algorithm	Type	Rationale
MultinomialNB	Probabilistic	Fast baseline, good for sparse text data
LogisticRegression	Linear Classifier	Interpretable, probabilistic outputs
Linear S-V	SVM Variant	Superior margin-based classification

3.2 Performance Results

Accuracy Comparison:

- LinearSupport Vector Machine: 90.40% ★ BEST
- LogisticRegression: 89.40%
- MultinomialNB: 79.75%

Linear Support Vector Machine: 1,808 correct predictions out of 2,000 (90.40% accuracy). Strong performance on majority classes (0, 1, 3, 4) with 86-95% per-class accuracy. Moderate performance on minority classes 2 and 5 (79% and 68% respectively) due to limited training data.

```
Training LinearSupportVectorMachine
--- LinearSupportVectorMachine ---
Accuracy: 0.904
              precision    recall  f1-score

           0     0.9481    0.9432    0.9456
           1     0.9176    0.9295    0.9235
           2     0.7826    0.7925    0.7875
           3     0.8993    0.9091    0.9042
           4     0.8853    0.8616    0.8733
           5     0.7377    0.6818    0.7087

    accuracy                         0.9040
   macro avg     0.8618    0.8529    0.8571
weighted avg     0.9037    0.9040    0.9038
```

LogisticRegression: 1,788 correct predictions (89.40% accuracy). Good balance across classes with weighted F1-score of 0.9176.

```
Training LogisticRegression
--- LogisticRegression ---
Accuracy: 0.894
              precision    recall  f1-score

           0     0.9214    0.9484    0.9347
           1     0.8708    0.9698    0.9176
           2     0.8480    0.6667    0.7465
           3     0.9031    0.8473    0.8743
           4     0.9048    0.8482    0.8756
           5     0.9714    0.5152    0.6733

    accuracy                         0.8940
   macro avg     0.9032    0.7992    0.8370
weighted avg     0.8953    0.8940    0.8902
```

MultinomialNB: 1,595 correct predictions (79.75% accuracy). Poor performance on minority classes despite high recall, indicating precision-recall imbalance.

```
Training MultinomialNB
--- MultinomialNB ---
Accuracy: 0.7975
              precision    recall  f1-score

           0     0.8018    0.9398    0.8653
           1     0.7374    0.9899    0.8452
           2     1.0000    0.2453    0.3939
           3     0.9497    0.6182    0.7489
           4     0.9006    0.6473    0.7532
           5     1.0000    0.1061    0.1918

    accuracy                         0.7975
   macro avg     0.8983    0.5911    0.6331
weighted avg     0.8331    0.7975    0.7701
```

3.3 Confusion Matrix Analysis

LinearSVC demonstrates:

- Strongest Performance: Classes 0 and 1 (majority classes) with >94% accuracy
- Moderate Performance: Classes 3 and 4 (medium-sized) with ~87-90% accuracy
- Challenging Cases: Classes 2 and 5 (minority classes) with limited training samples (79% and 68% accuracy)

4 Key Findings

Model Superiority: LinearSV outperforms both Naive Bayes and Logistic Regression due to optimal margin-based separation in high-dimensional TF-IDF space

Class Imbalance Challenge: Minority classes (2, 5) show degraded performance; larger training sets would improve generalization

Consistency: LinearSVC maintains balanced precision-recall across majority classes, supporting production deployment

Feature Quality: TF-IDF representation effectively captures discriminative vocabulary (unigrams + bigrams)

5 CONCLUSION

The text classification pipeline successfully achieved **90.40% accuracy using Linear Support Vector Machine**, demonstrating robust performance on the 6-class classification task. While class imbalance presents challenges for minority categories, the overall model quality supports practical deployment with ongoing monitoring and refinement.

PLOT DATA RESULTS:





